Why Data Cleaning Matters

Methods and Tools in R

Cagan Goksay
Fresenius University of Applied Sciences

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```
library(tidyverse)
my_data <- read.csv(“ASSIST_DATA.csv”)
md <- my_data
```

Introduction

Data cleaning is an essential step in data analysis.

Project Context

This project uses a Bundesliga 2023/24 football dataset.

Dataset Overview

- Rank
- Player
- Team
- Country
- Expected Assists
- Actual Assists
- Minutes Played
- Matches Played

First Look at the Data

head(md)

Last Rows of the Dataset

```
tail(md)
```

Understanding Data Structure

glimpse(md)

Variable Type Inspection

class(md$Country)

Unique Values

unique(md$Country)

Filtering Relevant Data

md |> select(Player, Team, Country) |> filter(Country == “GER” & Team == “Borussia Dortmund”) |> arrange(Player)

Descriptive Statistics

```
mean(md$Actual.Assists, na.rm = TRUE)
```

Detecting Missing Values

```
md |> select(Rank, Player, Expected.Assists, Actual.Assists)
|> filter(!complete.cases(.))
```

Removing Missing Values

```
md_without_na <- md |> select(Rank, Team,
Expected.Assists, Actual.Assists) |> na.omit()
```

```
head(md_without_na)
```

Cleaning Missing Values

md_clean <- md |> mutate(Team = ifelse(is.na(Team) | Team == “” | Team == ” “,”none”, Team), Expected.Assists = ifelse(is.na(Expected.Assists), 0, Expected.Assists), Actual.Assists = ifelse(is.na(Actual.Assists), 0, Actual.Assists))

head(md_clean)

Detecting Duplicate Players

```
md_duplicates <- md |> filter(duplicated(Player) | duplicated(Player, fromLast = TRUE)) |> arrange(Player)
```

```
md_duplicates |> select(Rank, Player, Team, Expected.Assists, Actual.Assists)
```

Visualization

ggplot(md_clean, aes(x = Expected.Assists, y = Actual.Assists)) + geom_point() + labs(title = “Expected Assists vs Actual Assists”, x = “Expected Assists”, y = “Actual Assists”)

Key Takeaways

- Data cleaning improves accuracy
- Missing values should be checked carefully
- Duplicate rows may distort results
- Clean datasets improve reliability

Conclusion

This project demonstrates how R and tidyverse tools support data cleaning and preprocessing workflows.